

Applied Assignment: Regression Analysis Research Project

Objective

Apply concepts of correlation and multiple linear regression to investigate a research question using real-world data and communicate findings in a professional research report.

Part 1. Research Methodology

1. Select a Dataset

Choose an existing dataset suitable for quantitative analysis (e.g., GSS, ANES, World Values Survey, or a dataset provided on the course webpage). The dataset must include:

- One continuous dependent variable (outcome).
- At least two independent variables (predictors).
- A mix of continuous and categorical predictors, if possible.

2. Research Question (RQ)

Formulate a clear, specific, and testable research question examining relationships among your variables.

Example:

Under what conditions does education level predict income across gender? (i.e., Does the relationship between education and income differ by gender?)

Your RQ should:

- Identify the outcome variable.
- Identify key predictors.
- Indicate the relationship being tested.

3. Hypotheses

State two directional, testable hypotheses derived from your research question.

- **Hypothesis 1 (H1):** Predict the direction of the relationship between your primary predictor and the outcome.
- **Hypothesis 2 (H2):** Predict the effect or direction of a secondary predictor or control variable.

Hypotheses must be stated in measurable terms.

Part 2. Methodology and Analysis Report

Prepare a report using the following sections.

1. Data and Methods

Data Source

Include a **FULL** description of the data. Describe the dataset, including:

- Dataset name
- Year(s) collected
- Data source
- Sampling method used in data collection

Sample

Report:

- Final analytic sample size (N)
- Number of observations removed and reasons for removal (e.g., missing data).

Measures

- **Dependent Variable (DV):** Define the outcome variable, level of measurement, and any transformations used.
- **Independent Variables (IVs):** Define your primary predictors.
- **Control Variables:** Identify variables included to account for potential confounding influences.

Data Preparation and Coding

Clearly describe all data preparation steps:

- Category collapsing or regrouping
- Dummy coding procedures
- Reverse coding
- Handling missing values
- Any other transformations applied

Explain the logic behind these decisions.

2. Statistical Analysis Plan

Describe how hypotheses were tested, including:

- Statistical software used (e.g., R, JASP, SPSS).
- Type of regression model estimated (e.g., OLS multiple linear regression).
- Diagnostic and assumption checks performed, such as:
 - Multicollinearity (e.g., VIF)
 - Residual analysis
 - Homoscedasticity
 - Linearity
 - Normality of residuals

3. Results

Descriptive Statistics

Provide a professional table summarizing:

- Means, standard deviations, and ranges for continuous variables.
- Frequencies or proportions for categorical variables.

Regression Results

Include:

- A professional regression table reporting B, SE, t, p, and standardized coefficients (β).
- At least one figure illustrating model results or relationships.

Narrative Interpretation

In clear prose:

- Report whether the overall model was statistically significant.

- Report model fit (R^2).
- Identify statistically significant predictors.
- Interpret coefficients substantively (e.g., “A one-unit increase in X is associated with a ____ change in Y”).
- State whether hypotheses were supported.
- **Summarize the overall findings in one concise paragraph addressing the research question.**

Submission Requirements

Submit a single document (PDF or Word) containing:

- Research methodology
- Methodology and analysis report
- Tables and figures

Include statistical code or software output as an appendix.